

What are existing open source and free AI literacy curriculums for primary, secondary and high schools? Which countries, language and age group they cover? what knowledge, competences, skills? How many cover Ethical AI and risks & benefits of AI?

Ten open source or free AI literacy curricula for K-12 education have been identified, with coverage across all school levels (primary through high school), originating primarily from the US and Spanish-speaking countries, teaching machine learning and programming skills through hands-on methods, and universally including AI ethics content with most addressing bias, societal impacts, and AI risks.

Abstract

Ten open source or free AI literacy curricula for school learners appear in the literature. The studies report curricula designed for primary, middle, and high school students. Specifically, three curricula target primary students, six target middle schoolers, and six target high school students—with some spanning multiple levels. Four curricula originate in the United States, three in Spain (or Spanish-speaking contexts), one in Australia, one in Europe, and two provide global or unspecified geographic coverage. Six curricula use English, two use Spanish, and two do not specify a primary language. One curriculum (Hollands & Breazeal, 2024) is explicitly available under a Creative Commons license; several others imply open or free status, while five do not mention it.

The curricula address technical and conceptual content that includes machine learning (in five studies) and programming using tools such as Scratch or AppInventor (in three studies). They integrate hands-on projects, workshops, and group activities to develop practical competencies alongside conceptual learning. All studies describe coverage of AI ethics—nine curricula include content on ethical use, bias, and societal impact—while seven address risks such as misinformation, bias, or legal and sustainability concerns. These findings delineate the current landscape of AI literacy resources for primary to high school education.

Paper search

Using your research question "What are existing open source and free AI literacy curriculums for primary, secondary and high schools? Which countries, language and age group they cover? what knowledge, competences, skills? How many cover Ethical AI and risks & benefits of AI?", we searched across over 126 million academic papers from the Semantic Scholar corpus. We retrieved the 50 papers most relevant to the query.

Screening

We screened in sources that met these criteria:

- **Target Population:** Does the study describe AI literacy curricula specifically designed for primary, secondary, or high school students (ages 5-18)?
- **Curriculum Accessibility:** Are the curricula explicitly described as open source, free, or freely available?
- **Curriculum Detail:** Does the study provide detailed descriptions of curriculum content, learning objectives, or educational components?
- **Implementation Status:** Does the study report on curricula that have been implemented, piloted, or fully developed (rather than purely theoretical)?

- **Educational Level Scope:** Is the curriculum NOT designed exclusively for higher education (university/college level) or adult learners?
- **Commercial Status:** Is the curriculum NOT a commercial or proprietary AI education program requiring payment or licensing fees?
- **AI Focus Specificity:** Is this NOT a general technology or computer science curriculum that only mentions AI as a minor component?
- **Practical Implementation:** Is this NOT purely a theoretical framework or conceptual paper without describing specific curriculum implementations?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Target Age Group and Educational Level:**

Identify the specific age range or educational level (e.g., primary, middle school, high school) for which the AI literacy curriculum is designed.

- Look in the introduction, methods, or participant description sections
- If multiple age groups are mentioned, list all
- If an exact age range is not specified, use the educational level as a proxy
- Format: e.g., "High school students (ages 14-18)" or "Middle school (grades 6-8)"
- If no clear age group is specified, write "Not specified"

- **Curriculum Coverage of AI Topics:**

Identify and list the specific AI knowledge, competencies, and skills covered in the curriculum:

- Scan the methods, curriculum description, and results sections
- List all topics mentioned (e.g., AI basics, machine learning, conversational AI, AI ethics)
- Pay special attention to ethical AI and AI risks/benefits
- Use direct quotes where possible
- If topics are not clearly specified, write "Not fully described"

- **Geographic and Language Coverage:**

Determine the country or region where the curriculum was developed or tested:

- Check methods section, participant demographics, and affiliations
- Note the primary language of instruction
- If multiple countries are involved, list all
- If no geographic information is provided, write "Not specified"
- Format: Country/Region, Primary Language

- **Curriculum Delivery Method:**

Identify how the AI literacy curriculum was delivered:

- Look for details about instructional approach in methods or implementation sections
- Note specific methods (e.g., online workshop, in-person class, project-based learning)

- Indicate if multiple delivery methods were used
- If delivery method is not clear, write "Not specified"
- Include any unique pedagogical approaches mentioned

- **Accessibility and Inclusivity Features:**

Identify any specific efforts to make the curriculum accessible or inclusive:

- Search for mentions of reducing barriers to participation
- Note any strategies to include underrepresented groups
- Look for mentions of minimal prerequisite knowledge or technology requirements
- If no specific accessibility features are mentioned, write "Not described"
- Use direct quotes if possible to capture the approach

- **Sample Size and Participant Characteristics:**

Extract information about the study participants:

- Total number of participants
- Breakdown of participants (e.g., students, teachers)
- Any demographic information provided
- If multiple groups were involved (e.g., students and teachers), list separately
- If information is incomplete, note what is missing
- Format: Total n = X (students: Y, teachers: Z)

Results

Characteristics of Included Studies

Curriculum Name/Program	Target Age Groups	Geographic Coverage	Open Source Status
Williams et al., 2022: Creative AI, Dancing with AI, How to Train Your Robot	Middle school students (ages 11–13)	United States, English	Free online resources; open or free status was implied in the paper, but not explicitly stated
Bellas et al., 2022: AI+ Erasmus+ Curriculum	High school students (ages 15–18)	Europe (Lithuania, Finland, Slovenia, Italy, Spain), English	Openly available; open or free status was implied
Judd, 2020: AI4ALL Open Learning	High school students	Not specified	Free and approachable; open or free status was implied
Banks & Renzella, 2025: Day of AI Australia	Upper primary and early secondary (ages 10–16)	Australia, English	No mention found
Destéfano et al., 2024: CreaTIC Academy Humanistic AI	Middle school students (ages 11–14)	Spain, Spanish; Argentina, Spanish	Free tools used; open or free status was implied

Curriculum Name/Program	Target Age Groups	Geographic Coverage	Open Source Status
Hollands & Breazeal, 2024: MIT/i2Learning Modular AI Literacy	Students aged 5–18 (primary to high school)	United States, English; global implementation	Freely downloadable/adaptable under Creative Commons
Van Brummelen et al., 2020: Conversational AI Curriculum	High school students (grades 8–12)	United States, English	No mention found
Zhang et al., 2022: DAILY Workshop	Middle school students (grades 7–9)	United States, English	No mention found
Moreno-León et al., 2024: EPCIA Project	Primary and secondary (grades 4–8)	Spain, Spanish	No mention found
Ng et al., 2023: Systematic Review	Secondary school students (junior and senior)	Not specified (global review)	No mention found

Target Age Groups:

- Primary school (ages ~5–10):3 curricula (Banks & Renzella, Hollands & Breazeal, Moreno-León et al.)
- Middle school (ages ~11–14):7 curricula (Williams et al., Banks & Renzella, Destéfano et al., Hollands & Breazeal, Zhang et al., Moreno-León et al.)
- High school (ages ~15–18):6 curricula (Bellas et al., Judd, Banks & Renzella, Hollands & Breazeal, Van Brummelen et al., Ng et al.)
- Multiple age groups:Some curricula spanned multiple age groups.
- We didn't find mention of any studies where the target age group was not specified.

Geographic Coverage:

- United States:4 curricula
- Australia:1 curriculum
- Spain:3 curricula; Argentina:1 curriculum
- Europe (Lithuania, Finland, Slovenia, Italy, Spain):1 curriculum
- English language:6 curricula; Spanish language:2 curricula
- Global or unspecified coverage:2 studies
- We didn't find mention of geographic coverage for 2 studies.

Open Source Status:

- Explicitly open/free (Creative Commons license):1 curriculum
- Open or free status implied:4 curricula
- We didn't find mention of open or free status for 5 curricula.

Thematic Analysis

Geographic Distribution and Language Coverage

Curriculum Name/Program	Country/Region(s)	Primary Language(s)
Williams et al., 2022	United States	English
Bellas et al., 2022	Europe (Lithuania, Finland, Slovenia, Italy, Spain)	English
Judd, 2020	Not specified in abstract	Not specified in abstract
Banks & Renzella, 2025	Australia	English
Destéfano et al., 2024	Spain; Argentina	Spanish
Hollands & Breazeal, 2024	United States; global implementation	English
Van Brummelen et al., 2020	United States	English
Zhang et al., 2022	United States	English
Moreno-León et al., 2024	Spain	Spanish
Ng et al., 2023	Not specified (global review)	Not specified in abstract

Geographic distribution:

- United States:4 studies
- Spain:3 studies
- Lithuania, Finland, Slovenia, Italy, Argentina, Australia:1 study each
- Global implementation or review:2 studies
- We didn't find mention of the country or region for 2 studies.

Primary language:

- English:6 studies
- Spanish:2 studies
- We didn't find mention of the primary language for 2 studies.

Summary:

- Most studies were conducted in English-speaking countries or used English as the primary language.
- Studies from Spanish-speaking countries and in Spanish were also present.
- We didn't find mention of any studies specifying more than one primary language.

Age Group Targeting and Educational Level Alignment

Curriculum Name/Program	Target Age Groups / Educational Level
Williams et al., 2022	Middle school (ages 11–13)
Bellas et al., 2022	High school (ages 15–18)
Judd, 2020	High school
Banks & Renzella, 2025	Upper primary and early secondary (ages 10–16)
Destéfano et al., 2024	Middle school (ages 11–14)
Hollands & Breazeal, 2024	Primary to high school (ages 5–18)
Van Brummelen et al., 2020	High school (grades 8–12)
Zhang et al., 2022	Middle school (grades 7–9)

Curriculum Name/Program	Target Age Groups / Educational Level
Moreno-León et al., 2024	Primary and secondary (grades 4–8)
Ng et al., 2023	Secondary school (junior and senior)

Educational levels and age groups:

- Primary (elementary) level students:3 studies (Banks & Renzella, Hollands & Breazeal, Moreno-León et al.)
- Middle school level students or equivalent ages/grades:6 studies (Williams et al., Banks & Renzella, Destéfano et al., Hollands & Breazeal, Zhang et al., Moreno-León et al.)
- High school level students or equivalent ages/grades:6 studies (Bellás et al., Judd, Banks & Renzella, Hollands & Breazeal, Van Brummelen et al., Ng et al.)
- Multiple educational levels (primary and secondary):3 studies (Banks & Renzella, Hollands & Breazeal, Moreno-León et al.)
- We didn't find mention of studies that targeted only early primary (below grade 4 or age 10), nor did we find mention of any that focused exclusively on post-secondary or adult learners.

Knowledge, Competencies, and Skills Framework

Curriculum Name/Program	Technical Skills	Conceptual Knowledge	Practical Competencies	Assessment Methods
Williams et al., 2022	Artificial intelligence basics, machine learning, generative adversarial networks, robotics	Perception, reasoning, learning, societal impact	Hands-on projects, critique of artificial intelligence artifacts	Projects, presentations, group activities
Bellás et al., 2022	Programming, embedded intelligence	Perception, actuation, reasoning, learning, ethics	Real-world problem solving, teamwork	Feedback, practical workshops
Judd, 2020	No mention found	Artificial intelligence basics, everyday applications, ethics	Discussion, community engagement	No mention found
Banks & Renzella, 2025	Machine learning, natural language processing	Artificial intelligence ethics	No mention found	No mention found
Destéfano et al., 2024	Coding (Scratch, AppInventor), machine learning for kids	Human cognition, philosophy of mind, ethics	Project-based, social research, critical reflection	No mention found

Curriculum Name/Program	Technical Skills	Conceptual Knowledge	Practical Competencies	Assessment Methods
Hollands & Breazeal, 2024	Machine learning, data science, artificial intelligence in social media, ChatGPT	Artificial intelligence vocabulary, mechanics, societal impact, ethics	Hands-on activities, design challenges	Teacher/student feedback, knowledge tests
Van Brummelen et al., 2020	Conversational artificial intelligence, machine learning, Alexa skills	Big 5 artificial intelligence ideas, ethics, data literacy	Design, testing, future artificial intelligence planning	Tutorials, questionnaires, feedback
Zhang et al., 2022	Machine learning, logic systems, generative adversarial networks, Teachable Machine	Bias, societal impacts, career futures	Interactive, hands-on, career exploration	Pre/post tests, qualitative feedback
Moreno-León et al., 2024	Programming (Scratch, machine learning for kids), machine learning models	Artificial intelligence basics, societal impact	Project-based, demystifying artificial intelligence	Quantitative tests, questionnaires
Ng et al., 2023	No mention found	Ethical use, basic to advanced artificial intelligence concepts	Collaborative, interdisciplinary problem-solving	Knowledge tests, questionnaires, qualitative analysis

Technical skills:

- Machine learning:5 studies
- Programming or coding (including Scratch, AppInventor, machine learning for kids):3 studies
- Generative adversarial networks:2 studies
- Other technical skills (robotics, embedded intelligence, natural language processing, data science, artificial intelligence in social media, ChatGPT, conversational artificial intelligence, Alexa skills, logic systems, Teachable Machine, machine learning models):Each found in 1 study
- We didn't find mention of a full description of technical skills in 2 studies.

Conceptual knowledge:

- Ethics or ethical use:6 studies
- Societal impact or societal impacts:4 studies
- Perception, reasoning, learning:Each found in 2 studies
- Artificial intelligence basics:2 studies
- Other conceptual topics (human cognition, philosophy of mind, artificial intelligence vocabulary, mechanics, Big 5 artificial intelligence ideas, data literacy, bias, career futures, everyday applications, basic to advanced

artificial intelligence concepts):Each found in 1 study

Practical competencies:

- Hands-on or interactive activities/projects:4 studies
- Project-based learning:2 studies
- Other competencies (real-world problem solving, teamwork, critique of artificial intelligence artifacts, discussion, community engagement, social research, critical reflection, design/design challenges, testing, future artificial intelligence planning, career exploration, demystifying artificial intelligence, collaborative/interdisciplinary problem-solving):Each found in 1 study
- We didn't find mention of practical competencies in 2 studies.

Assessment methods:

- Knowledge or quantitative tests:4 studies
- Questionnaires:3 studies
- Feedback:3 studies
- Projects, presentations, group activities, practical workshops, tutorials, qualitative feedback or analysis:Each found in 1 study
- We didn't find mention of assessment methods in 4 studies.

Ethical Artificial Intelligence Education and Risk-Benefit Coverage

Curriculum Name/Program	Ethical Artificial Intelligence Components	Risk Coverage	Benefit Coverage	Integration Approach
Williams et al., 2022	Artificial intelligence ethics, societal implications, stakeholder analysis	Deepfakes, misinformation, bias	Societal impact, positive/negative effects	Embedded in projects and discussions
Bellas et al., 2022	Sustainability, ethics, legal aspects (sustainability, ethics, legal)	Ethics, legal, sustainability	No mention found	Dedicated teaching units
Judd, 2020	Ethics, social impacts	No mention found	Future uses of artificial intelligence	Discussion-based, integrated
Banks & Renzella, 2025	Artificial intelligence ethics	No mention found	No mention found	Included as core concept
Destéfano et al., 2024	Artificial intelligence ethics, gender/skin-type bias, social sensitivity	Artificial intelligence risks, bias, social experiments	Universal topics, social good	Humanistic, project-based, reflective

Curriculum Name/Program	Ethical Artificial Intelligence Components	Risk Coverage	Benefit Coverage	Integration Approach
Hollands & Breazeal, 2024	Responsible, equitable, ethical use, algorithmic bias	Benefits/harms, bias	Societal benefits, optimism about artificial intelligence	Modular, hands-on, adaptable
Van Brummelen et al., 2020	Artificial intelligence ethics, bias, strengths/weaknesses	Bias, hyper-realistic text/speech generation	Positive/negative effects of conversational agents	Dedicated modules, project reflection
Zhang et al., 2022	Bias, ethical considerations, societal impacts	Bias, mitigation strategies	Career futures, adaptability	Integrated throughout curriculum
Moreno-León et al., 2024	Societal impact, demystifying artificial intelligence	Positive/negative effects	Societal impact	Project-based, implicit
Ng et al., 2023	Ethical use, ethical dimensions	No mention found	No mention found	Collaborative, interdisciplinary

Ethical artificial intelligence components:

- Explicit coverage of ethics, artificial intelligence ethics, or ethical use/dimensions:9 studies
- Bias (including algorithmic, gender, or skin-type bias):6 studies
- Societal implications or impacts:3 studies
- Social impacts or social sensitivity:2 studies
- Other components (responsible/equitable use, legal aspects, sustainability, stakeholder analysis, strengths/weaknesses, demystifying artificial intelligence):Each found in 1 study

Risk coverage:

- Bias:7 studies
- Ethics or ethical risks:2 studies
- Legal and sustainability risks:Each found in 1 study
- Other risks (deepfakes, misinformation, artificial intelligence risks, social experiments, benefits/harms, hyper-realistic text/speech generation, mitigation strategies, positive/negative effects):Each found in 1 study
- We didn't find mention of risk coverage in 3 studies.

Benefit coverage:

- Societal impact or societal benefits:3 studies
- Positive/negative effects:2 studies
- Future uses, career futures, or adaptability:3 studies
- Social good, universal topics, optimism about artificial intelligence, conversational agents:Each found in 1 study
- We didn't find mention of benefit coverage in 3 studies.

Integration approach:

- Project-based or project reflection approaches:3 studies
- Discussion-based approaches:2 studies
- Embedded or integrated approaches (including integration throughout the curriculum):3 studies
- Dedicated teaching units or modules:2 studies
- Other approaches (humanistic, reflective, modular, hands-on, adaptable, core concept, collaborative/interdisciplinary, implicit):Each found in 1 study

: Source: Williams et al., 2022 : Source: Williams et al., 2022 : Source: Bellas et al., 2022 : Source: Bellas et al., 2022 : Source: Judd, 2020 : Source: Judd, 2020 : Source: Banks & Renzella, 2025 : Source: Banks & Renzella, 2025 : Source: Destéfano et al., 2024 : Source: Destéfano et al., 2024 : Source: Hollands & Breazeal, 2024 : Source: Hollands & Breazeal, 2024 : Source: Van Brummelen et al., 2020 : Source: Van Brummelen et al., 2020 : Source: Zhang et al., 2022 : Source: Zhang et al., 2022 : Source: Moreno-León et al., 2024 : Source: Moreno-León et al., 2024 : Source: Ng et al., 2023 : Source: Ng et al., 2023

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